

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 15-01 NRC ESG-1

SCENARIO NUMBER: 15-01 NRC ESG-1

EFFECTIVE DATE: See Approval Dates

EXPECTED DURATION: 60 minutes

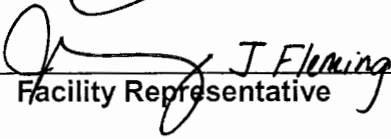
REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary
New issue for 15-01 ILOT NRC exam

PREPARED BY: G Gauding 7-18-16
Lead Regulatory Exam Author Date

APPROVED BY:  10/11/16
Operations Training Manager Date

APPROVED BY:  10/11/16
Facility Representative Date

I. OBJECTIVES

- A. Given the order or indications of a charging system malfunction, perform actions as the nuclear control operator to RESPOND to the loss or malfunction of the charging system in accordance with S2.OP-AB.CVC-0001.
- B. Given indication of a loss or malfunction of the Charging system DIRECT corrective action for a Charging System malfunction in accordance with S2.OP-AB.CVC-0001.
- C. Given the indications of a reactor coolant system (RCS) malfunction or leak, perform actions as the nuclear control operator to RESPOND to the malfunctioning accordance with S2.OP-AB.RC-0001.
- D. Given the indications of a reactor coolant system (RCS) malfunction or leak, DIRECT the response to the malfunction in accordance with S2.OP-AB.RC-0001.
- E. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with 2-EOP-TRIP-1.
- F. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with 2-EOP-TRIP-1.
- G. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- H. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.
- I. Given a safety injection has occurred and inadequate ECCS flow exists, ALIGN equipment to obtain required ECCS flow in accordance with station procedures.
- J. Given the order or indications of a loss of coolant accident (LOCA), complete actions as the nuclear control operator to PERFORM the immediate response to the LOCA in accordance with the approved station procedures.
- K. Given indication of a loss of coolant accident (LOCA), DIRECT the immediate response to the LOCA in accordance with the approved station procedures.
- L. Given conditions that allow SI termination, perform actions as the nuclear control operator to TERMINATE safety injection in accordance with the approved station procedures.
- M. Given conditions that allow SI termination, DIRECT the response to the safety injection termination in accordance with the approved station procedures.

II. MAJOR EVENTS

- A. Controlling PZR Level Channel fails low
- B. 20 gpm RCS leak
- C. 150 gpm SBLOCA
- D. SBLOCA escalation during SI termination

III. SCENARIO SUMMARY

- A. The crew will take the watch with the unit at 100% power, MOL. PZR Pressure Channel III is selected for Control.
- B. After the crew assumes the watch, the controlling PZR level channel will fail low. CVCS Letdown will isolate, and the Master Flow Controller will raise charging flow. The crew will enter S2.OP-AB.CVC-0001, Loss of Charging. The crew will take manual control of charging flow and return PZR level to program. The crew will remove the failed channel from service. The crew will restore normal letdown. The CRS will identify Tech Specs.
- C. After the PZR level channel failure has been addressed, a 20 gpm RCS leak will occur. The crew will diagnose the leak. The crew will swap to a centrifugal charging pump and a 45 gpm orifice, and estimate the leak rate. The CRS will identify Tech Specs.
- D. The CRS will determine a unit shutdown is required based on the RCS leak. The crew will initiate a controlled shutdown.
- E. After the downpower has commenced, the RCS leak rate will rise to 150 gpm. The crew will diagnose deteriorating primary conditions. When it is determined that PZR level will be unable to be maintained >11%, the crew will take the CAS action to trip the reactor and initiate Safety Injection.
- F. The crew will perform plant stabilization and diagnostics in EOP-TRIP-1, Reactor Trip or Safety Injection. The crew will identify no AFW flow. 2A SEC will not reset, and will require deenergization to start 21 AFW pump. The crew will establish at least 22E4 lbm/hr AFW flow. (CT#1)
- G. The crew will determine there is no High Head ECCS flow, and open the BIT Isolation Valves to establish High Head ECCS Injection (CT#2).
- H. The crew will transition to EOP-LOCA-1, Loss of Reactor Coolant when directed in TRIP-1.
- I. The crew will reset Safeguards in LOCA-1, and determine SI termination criteria are met to transition to EOP-TRIP-3, SI Termination.
- J. The crew will stop all but one Charging pump and align CVCS for normal charging. When evaluating the leak, the SBLOCA will get significantly larger. The crew will diagnose degrading conditions and take the CAS action to start ECCS pumps as necessary, and go to EOP-LOCA-2.
- K. The scenario will terminate when the transition to LOCA-2 has been made.

IV. INITIAL CONDITIONS

___ IC-231

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
___ 1	VC1and VC4 C/T
___ 2	RCPs (SELF CHECK)
___ 3	RTBs (SELF CHECK)
___ 4	MS167s (SELF CHECK)
___ 5	500 KV SWYD (SELF CHECK)
___ 6	SGFP Trip (SELF CHECK)
___ 7	23 CV PP (SELF CHECK)
___ 8	PZR Pressure Channel III selected for Control.
___ 9	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
		EVENT ACTION: COMMAND: PURPOSE:

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
___ 01	PR0017A PZR LEVEL CH I (LT459) FAILS H/L	N/A	N/A	N/A	RT-1	0
___ 02	RC0002 RCS LEAK INTO CONTAINMENT (equiv to 0-4 inches)	N/A	N/A	N/A	RT-3	20
___ 03	RP0342 2SJ12 FAILS TO AUTO OPEN ON SEC	N/A	N/A	N/A	N/A	
___ 04	RP0343 2SJ13 FAILS TO AUTO OPEN ON SEC	N/A	N/A	N/A	N/A	
___ 05	RP318E1 Aux Feed Pump 21 Fails to Start on SEC	N/A	N/A	N/A	N/A	
___ 06	RP318E2 Aux Feed Pump 22 Fails to Start on SEC	N/A	N/A	N/A	N/A	
___ 07	AF0353A 21 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	
___ 08	AF0353B 22 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
___ 01	DG01D DEENERGIZE "A" SEC CABINET	N/A	N/A	N/A	RT-9	YES
___ 02	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10
___ 03	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160
___ 04	MS05A 21MS45 21 STM GEN STM SUP-23 AFP	N/A	N/A	N/A	N/A	0
___ 05	MS06A 23MS45 23 STM GEN STM SUP-23 AFP	N/A	N/A	N/A	N/A	0

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
___ 01	CB09 A DI KCB09ASE 2A SFGD EMER OPR-EMERG LOADING RESET	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

Description

1.

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Controlling PZR Level Channel Fails LOW Simulator Operator: Insert RT-1 on direction from Lead Evaluator. MALF: PR0017A PZR LVL CH I (LT459) FAILS H/L Severity: 0			
	RO reports unexpected OHA E-36, PZR HTR OFF LVL LO.		
	RO diagnoses controlling PZR level channel failure.		
	RO may request permission to place the Master Flow Controller in manual prior to direction in AB.CVC to do so.		
	CRS enters S2.OP-AB.CVC-0001, Loss of Charging.		
	CRS directs initiation of Attachment 1 CAS.		
	RO reports charging pumps are running, and no indication of cavitation.		
	RO reports PZR level channel I has failed low.		
	If not previously performed, RO takes manual control of charging and maintains PZR level within band assigned by CRS.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Proceed to next event after Tech Spec call at direction from Lead evaluator.	RO selects operable channel for Control, Alarm, and Recorders.		
	RO restores PZR heaters to pre-event condition.		
	RO announces OHA E-20 PZR HTR ON LVL HI as expected when it annunciates due to no letdown with charging flow established.		
	PO restores letdown by: - Ensuring open 2CV7 - Placing 2CV18 in manual and closing, then opening until close PB extinguishes. - Opening 2CV2 and 2CV277, then placing both in auto. - Ensuring charging flow is ~85-90 gpm. - Opening a 75 gpm orifice isolation valve while adjusting 2CV18 to control letdown pressure ~300 psig. - Placing 2CV18 in auto.		
	RO returns Master Flow Controller to auto when PZR level returns to program.		
	CRS contacts IC for assistance in removing channel from service.		
	CRS enter LCO 3.3.1.1 action 6.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
3. 20 gpm RCS leak			
Simulator Operator: Insert RT-3 on direction from Lead Evaluator			
MALF: RC0002 – RCS Leak into Containment. Final Value: 20 gpm			
	RO reports that charging flow is rising (if MFC has been returned to auto) and PZR level is lowering slowly.		
	Crew reports reading on 2R11A containment radiation monitor is rising.		
	RO reports unexpected OHA C-2 CNTMT SUMP PMP START when it occurs.		
	CRS enters S2.OP-AB.RC-0001, Reactor Coolant System Leak.		
	CRS directs initiation of Attachment 1 CAS.		
	RO reports RCS temperature > 350°F in Mode 1.		
	RO reports PZR level is lowering.		
Note: PZR level is above program, and charging flow will be lower than normal to reduce PZR level. If charging flow is raised here to stabilize level, then step to place a centrifugal charging pump in			

Role Play: If contacted state Rad Pro recommends placing 2 CFCU's in Low Speed and 2 CFCU's in High Speed.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Role Play: After RCS leakage Tech Spec has been identified, if CRS does not initiate a power reduction to bring the unit offline, call Control Room as Operations Director and instruct the CRS to initiate a 30% per hour load reduction to 20% in preparation for taking Unit 2 offline to investigate and repair the RCS leak. CRS may enter S2.OP-AB.LOAD-0001 Rapid Load Reduction to perform the power reduction. Proceed to next event on direction from Lead Evaluator. 3. SBLOCA	CRS enters TSAS 3.4.7.2.b action b for RCS leakage, and 3.6.1.4 for containment pressure if it has reached 0.3 psig.		
	CRS briefs power reduction.		
	RO initiates boration.		
	PO initiates turbine load reduction at 30% per hour.		
Simulator Operator: MODIFY MALF RC0002 from 20 to 150 on direction from Lead Evaluator.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
4. AFW fails to initiate	RO reports PZR level trending down and adjusts charging flow to maintain PZR level.		
	RO reports PZR level will not be able to be maintained >17%.		
	CRS directs Rx trip and Safety Injection		
	RO trips the reactor and confirms the reactor trip.		
	RO initiates Safety Injection.		
	RO continues Immediate Actions of TRIP-1 - Trips the Main Turbine - Reports at least one 4KV vital bus energized. - Reports SI initiated.		
	CRS and RO verify performance of immediate actions.		
	PO reports SEC loading is not complete for 2A and 2B vital buses, 21 and 22 AFW pumps have failed to start. Reports 23 AFW pump started but is not providing any AFW flow.		
	RO blocks 2A and 2B SECs.		
	PO resets 2A and 2B SECs.		
	PO reports 2A SEC will not reset.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS dispatches operator to deenergize 2A SEC.		
Simulator Operator: 2 minutes after being dispatched, insert RT-9 to deenergize 2A SEC.			
	RO starts 22 AFW pump.		
CT#1 (CT-5) Establish at least 22E4 lbm/hr AFW flow to the SG's prior to a transition to FRHS-1 being required. SAT _____ UNSAT _____			
	PO reports AFW pump status, and starts 21 AFW pump when 2A SEC is deenergized.		
	PO reports not all valve groups in Table B are in safeguards position.		
	PO reports 2SJ12 and 2SJ13 BIT isolation valves have not opened.		
	RO opens 2SJ12 and 2SJ13 and reports BIT flow is established.		
CT#2 (CT-6) Establish flow from at least one charging pump prior to transition out of TRIP-1. SAT _____ UNSAT _____			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports 21 and 22CA330s are shut.		
	RO reports containment pressure has remained <15 psig.		
	PO reports no high steam flow condition.		
	CRS directs implementation of the ECG.		
	PO reports all 4KV vital buses energized.		
	RO reports CAV in AP mode.		
	RO reports 2 switchgear supply and one exhaust fan are running.		
	RO reports 2 or more CCW pumps running.		
	RO reports RCS not aligned for Cold Leg Recirc.		
	RO reports charging flow is at least 100 gpm on SI systems charging flow meter.		
	RO reports RCS pressure is not <1540 psig.		
	PO reports total AFW flow is 22E4 lbm/hr or 9% level in at least one SG NR level, then maintains 19-33% level.		
	RO reports all RCPs running, and RCS temperature is stable at or trending to 547°F.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: CFST monitoring will commence after the transition out of TRIP-1. STA will report to the control room 10 minutes after being paged.	RO reports both RTBs are open.		
	RO reports both PZR PORVs are shut with their Block Valves open.		
	RO reports all RCPs in service and PZR spray operating as expected for current plant conditions.		
	RO reports RCS pressure >1350 psig and maintains seal injection flow to all RCPs.		
	PO reports no indication of faulted SGs.		
	PO/RO report no indications of a steam generator tube rupture.		
	RO reports radiation monitor channel status in Table F.		
	PO reports 2 or more channels rising or in warning or alarm.		
	CRS transitions to EOP-LOCA-1, Loss of Reactor Coolant.		
	PO reports no faulted SGs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports total AFW flow is 22E4 lbm/hr or 9% level in at least one SG NR level, then maintains 19-33% level.		
	PO reports no ruptured SGs.		
	RO resets SI, Phase A, and Phase B isolations.		
	RO opens 21 and 22CA330s.		
	PO resets 2C SEC, and reports 230V control centers reset.		
	RO resets SGBD Sample Isolation Bypass and opens 21-24SS94s.		
	CRS directs chemistry to sample SGs for activity and boron.		
	RO reports PORVs are shut with Block valves open.		
	RO reports subcooling > 0°F.		
	PO reports AFW flow >22E4 lbm/hr or SG NR level >9%.		
	RO reports RCS pressure is stable or rising.		
	RO reports PZR level >11%.		
	CRS transitions to EOP-TRIP-3, SI Termination.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO resets SI, Phase A, and Phase B isolations if not previously reset.		
	RO opens 21 and 22CA330s if not previously performed.		
	RO reports spray valves are not failed open.		
	PO resets 2C SEC and reports 230V control centers reset if not previously performed.		
	RO stops all but one charging pump.		
	RO reports RCS pressure is stable or rising.		
	RO reports charging pump suction is aligned to the RWST.		
	RO opens 2CV139 and 2CV140 if shut.		
	RO shuts BIT isolation valves 2SJ4, 2SJ5, 2SJ12, 2SJ13.		
	RO shuts 2CV55 Charging Flow Control valve.		
	RO opens 2CV68 and 2CV69 Charging Discharge valves.		
Simulator Operator: MODIFY MALF RC0002 from 150 to 2000 when 2CV68 and 2CV69 are open.			
	RO adjusts 2CV55 to maintain PZR level		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Terminate scenario on direction from Lead Evaluator after the transition to LOCA-2 is made.	>25%.		
	RO reports PZR level is not stable or rising.		
	PO reports no faulted SGs.		
	RO opens BIT isolation valves 2SJ4,5,12,13, and shuts normal charging line isolation valves 2CV68 and 69.		
	CRS goes to EOP-LOCA-2, Post LOCA Cooldown and Depressurization.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0004, Power Operation
- F. S2.OP-AB.CVC-0001, Loss of Charging
- G. S2.OP-AB.RC-0001, Reactor Coolant System Leak
- H. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- I. 2-EOP-TRIP-3, Safety Injection Termination
- J. 2-EOP-LOCA-1, Loss of Reactor Coolant
- K. 2-EOP-LOCA-2, Post LOCA Cooldown and Depressurization

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 100% RCS BORON: 857 MWe 1220

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA
REACTIVITY PARAMETERS

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

ABNORMAL PLANT CONFIGURATIONS:

PZR Pressure Channel III is selected for Control

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 15-01 ILOT NRC ESG-1 **REVIEWER:** C Recchione

Initials Qualitative Attributes

- | | | |
|----|-----|---|
| CR | 1. | The scenario has clearly stated objectives in the scenario. |
| CR | 2. | The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events. |
| CR | 3. | The scenario consists mostly of related events. |
| CR | 4. | Each event description consists of: <ul style="list-style-type: none"> • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point |
| CR | 5. | No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event. |
| CR | 6. | The events are valid with regard to physics and thermodynamics. |
| CR | 7. | Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives. |
| CR | 8. | The simulator modeling is not altered. |
| CR | 9. | All crew competencies can be evaluated. |
| CR | 10. | The scenario has been validated. |
| CR | 11. | If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario. |
| CR | 12. | ESG-PSA Evaluation Form is completed for the scenario at the applicable facility. |

ATTACHMENT 4
SIMULATOR SCENARIO REVIEW CHECKLIST

Initial Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)

GG	3	Malfunctions after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

15-01 NRC ESG-1

CT#1 (CT-4) Establish the minimum required feed flow prior to a transition to FRHS-1 being required.

Basis: Failure to establish the required minimum feed flow rate under the postulated plant conditions, results in "adverse consequence(s) or significant degradation in the mitigative capability of the plant."

Failure to perform the critical task represents a "demonstrated inability of the crew to:

- Take an action or combination of actions that would prevent a challenge to plant safety
- Recognize a failure / incorrect auto actuation of an ESF system or component."

CT#2 (CT-6) Establish flow from at least one high-head ECCS pump before transition out of TRIP-1.

Basis: Failure to manually start at least one Charging pump under the postulated conditions constitutes misoperation or incorrect crew performance in which the crew does not prevent "degraded emergency core cooling system (ECCS) ... capacity." In this case, at least one Charging/SI pump (high-head SI pump for LP plants) can be manually started from the control room. Therefore, failure to manually start a Charging/SI pump (high-head SI pump for LP plants) also represents a failure by the crew to "demonstrate the following abilities:

- Effectively direct or manipulate engineered safety feature (ESF) controls that would prevent a significant reduction of safety margin beyond that irreparably introduced by the scenario
- Recognize a failure or an incorrect automatic actuation of an ESF system or component"

Additionally, under the postulated plant conditions, failure to manually start a Charging/SI pump (high-head SI pump for LP plants) (when it is possible to do so) is a "violation of the facility license condition."

Note to Evaluators: CT numbers in parentheses are the corresponding Westinghouse ERG Rev. 2- based Critical Task procedure WCAP-17711-NP

ATTACHMENT 6**ESG-PRA RELATIONSHIP EVALUATION****EVENTS LEADING TO CORE DAMAGE**

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
N	Steam Generator Tube Rupture	N	Loss of CCW
N	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
Y	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	Y	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.